SG No. 207 O^1 $P432$ [cubic]

* plus set: $+[0, 0, 0]$

Table 1: Wyckoff site: `1a`, site symmetry: `432`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `1b`, site symmetry: `432`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `3c`, site symmetry: `42.2`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,17,18,19,20]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,11,12,21,22,23,24]`

Table 4: Wyckoff site: `3d`, site symmetry: `42.2`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,3,4,17,18,19,20]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[5,6,7,8,13,14,15,16]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[9,10,11,12,21,22,23,24]`

Table 5: Wyckoff site: `6e`, site symmetry: `4..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,17,20]`
2	$[-x, 0, 0]$	`[2,3,18,19]`
3	$[0, x, 0]$	`[5,8,13,16]`
4	$[0, -x, 0]$	`[6,7,14,15]`
5	$[0, 0, x]$	`[9,12,22,23]`
6	$[0, 0, -x]$	`[10,11,21,24]`

Table 6: Wyckoff site: `6f`, site symmetry: `4..`

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,17,20]`
2	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,18,19]`
3	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[5,8,13,16]`
4	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[6,7,14,15]`

continued ...

Table 6

No.	position	mapping
5	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[9,12,22,23]
6	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[10,11,21,24]

Table 7: Wyckoff site: 8g, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,5,9]
2	$[-x, -x, x]$	[2,7,12]
3	$[-x, x, -x]$	[3,8,10]
4	$[x, -x, -x]$	[4,6,11]
5	$[x, x, -x]$	[13,17,21]
6	$[-x, -x, -x]$	[14,19,24]
7	$[x, -x, x]$	[15,20,22]
8	$[-x, x, x]$	[16,18,23]

Table 8: Wyckoff site: 12h, site symmetry: 2..

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,4]
2	$\left[-x, \frac{1}{2}, 0\right]$	[2,3]
3	$\left[0, x, \frac{1}{2}\right]$	[5,8]
4	$\left[0, -x, \frac{1}{2}\right]$	[6,7]
5	$\left[\frac{1}{2}, 0, x\right]$	[9,12]
6	$\left[\frac{1}{2}, 0, -x\right]$	[10,11]
7	$\left[\frac{1}{2}, x, 0\right]$	[13,16]
8	$\left[\frac{1}{2}, -x, 0\right]$	[14,15]
9	$\left[x, 0, \frac{1}{2}\right]$	[17,20]
10	$\left[-x, 0, \frac{1}{2}\right]$	[18,19]
11	$\left[0, \frac{1}{2}, -x\right]$	[21,24]
12	$\left[0, \frac{1}{2}, x\right]$	[22,23]

Table 9: Wyckoff site: 12i, site symmetry: ..2

No.	position	mapping
1	$[0, y, y]$	[1,18]
2	$[0, -y, y]$	[2,20]
3	$[0, y, -y]$	[3,17]
4	$[0, -y, -y]$	[4,19]
5	$[y, 0, y]$	[5,15]
6	$[y, 0, -y]$	[6,13]
7	$[-y, 0, y]$	[7,16]
8	$[-y, 0, -y]$	[8,14]
9	$[y, y, 0]$	[9,21]

continued ...

Table 9

No.	position	mapping
10	$[-y,\ y,\ 0]$	`[10,23]`
11	$[y,\ -y,\ 0]$	`[11,22]`
12	$[-y,\ -y,\ 0]$	`[12,24]`

Table 10: Wyckoff site: `12j`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ y\right]$	`[1,18]`
2	$\left[\frac{1}{2},\ -y,\ y\right]$	`[2,20]`
3	$\left[\frac{1}{2},\ y,\ -y\right]$	`[3,17]`
4	$\left[\frac{1}{2},\ -y,\ -y\right]$	`[4,19]`
5	$\left[y,\ \frac{1}{2},\ y\right]$	`[5,15]`
6	$\left[y,\ \frac{1}{2},\ -y\right]$	`[6,13]`
7	$\left[-y,\ \frac{1}{2},\ y\right]$	`[7,16]`
8	$\left[-y,\ \frac{1}{2},\ -y\right]$	`[8,14]`
9	$\left[y,\ y,\ \frac{1}{2}\right]$	`[9,21]`
10	$\left[-y,\ y,\ \frac{1}{2}\right]$	`[10,23]`
11	$\left[y,\ -y,\ \frac{1}{2}\right]$	`[11,22]`
12	$\left[-y,\ -y,\ \frac{1}{2}\right]$	`[12,24]`

Table 11: Wyckoff site: `24k`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`
4	$[x,\ -y,\ -z]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$[z,\ -x,\ -y]$	`[6]`
7	$[-z,\ -x,\ y]$	`[7]`
8	$[-z,\ x,\ -y]$	`[8]`
9	$[y,\ z,\ x]$	`[9]`
10	$[-y,\ z,\ -x]$	`[10]`
11	$[y,\ -z,\ -x]$	`[11]`
12	$[-y,\ -z,\ x]$	`[12]`
13	$[y,\ x,\ -z]$	`[13]`
14	$[-y,\ -x,\ -z]$	`[14]`
15	$[y,\ -x,\ z]$	`[15]`
16	$[-y,\ x,\ z]$	`[16]`
17	$[x,\ z,\ -y]$	`[17]`
18	$[-x,\ z,\ y]$	`[18]`
19	$[-x,\ -z,\ -y]$	`[19]`
20	$[x,\ -z,\ y]$	`[20]`
21	$[z,\ y,\ -x]$	`[21]`

continued ...

Table 11

No.	position	mapping
22	$[z, -y, x]$	[22]
23	$[-z, y, x]$	[23]
24	$[-z, -y, -x]$	[24]